

Minnesota Freight Industry · Deep-Dive Analysis

Prepared by JADE OS · Operations Workbench · June 07, 2026

1 · Executive Summary

Minnesota's freight market is a deep mid-market base anchored by the Twin Cities I-94/I-35 corridor, an active intermodal interchange, and a mature roster of 50-250 truck operators. It is structurally well-suited to an AI operating layer that sits ABOVE existing TMS/telematics and converts visibility into decisions.

- ~ 13,000 motor carriers based in MN (FMCSA registered) · mid-market dominant
- ATA estimates US freight tonnage trends remain positive but margin-compressed
- Driver wages, fuel, and insurance all rose faster than rates in 2023-24 (ATRI 2024)
- Mid-market TMS penetration high but agent-layer adoption still under 10%
- MN DOT freight plan emphasizes corridor reliability + intermodal investment
- Implication · the operator who composes agents on top of existing TMS wins the mid-market

2 · Fleet Size Distribution & Service Mix

Minnesota's carrier base skews mid-market. Public registrations + Trucking Association of Minnesota rosters give a workable distribution:

- Single-truck operators · 55-60% of registrants (FMCSA), small share of tonnage
- 2-24 trucks · ~ 28% of fleets
- 25-99 trucks · ~ 10-12% of fleets, meaningful tonnage share
- 100-499 trucks · ~ 3-4% of fleets, disproportionate tonnage and tech budget
- 500+ trucks · < 1%, but includes national-anchor carriers (ATS, Dart, C.H. Robinson asset-light)

Service mix · Truckload still dominant (60-70% of revenue), LTL 15-20%, intermodal 5-10% with growth (BNSF Northtown · CP Rail Pig's Eye), specialized/hazmat the rest. (Sources · ATA Trucking Industry Profile · MN DOT Freight Plan · TAM rosters)

3 · Growth Trends & Margin Pressure Drivers

Headline · revenue growth is positive but uneven; operating costs have outrun rate increases for two consecutive years. ATRI's 2024 Operational Costs of Trucking puts marginal cost per mile at \$2.27, with driver wages + benefits the single largest line item.

- Driver shortage · ATA still cites 60-80k driver shortfall industry-wide; turnover ~78% mid-market
- Fuel · ULSD volatility, EIA weekly retail diesel still elevated vs. 2019 baseline

- Insurance · premiums up 15-30% YoY for many mid-market carriers post-litigation environment
- Regulatory · FMCSA Compliance/Safety Accountability program continues to weight enforcement
- Implication · cost pressure forces operating-margin recovery via decisions, not just rates

4 · Technology Adoption Rates

Adoption is high for the basics, low for the next layer. The opportunity is between visibility and decisions.

- ELD penetration · ~100% for >24-hour interstate (FMCSA mandate since 2017)
- TMS penetration · ~70% for 25-99 truck range, ~95% for 100+
- Telematics (Geotab/Samsara/Omnitracs) · ~80% for 50+ trucks
- AI agent layer · single-digit penetration; the gap JADE OS targets
- Top TMS vendors operating in MN · Descartes, McLeod, TMW, Selerant, MercuryGate

5 · Buyer Personas · 4 Archetypes

Persona A · Small Regional · 25-75 trucks. Owner-operator-adjacent, dispatch-tribal, fuel-sensitive, thin IT bench. Pain · margin recovery, retention.

Persona B · Mid-Market · 100-250 trucks. Multi-dispatcher, multiple TMS modules, dedicated safety lead. Pain · coordination cost, audit prep, retention.

Persona C · Intermodal-Leaning · any size. Yard moves + dray, container dwell penalties, rail interchange timing. Pain · driver wait time, capacity precision.

Persona D · Specialized / Hazmat · 50-150 trucks. Premium insurance, regulatory burden, customer SLA strict. Pain · compliance load, claims exposure.

6 · Implications & Recommended Posture

An AI-agent vendor entering MN should:

- Lead with the Dispatch and Compliance agents (highest perceived pain × lowest integration risk)
- Position ABOVE existing TMS (Descartes/McLeod/TMW) rather than displacing it
- Ground every claim in MN-specific benchmark numbers (rate floor, fuel, retention)
- Structure pilots in 90-day cycles with success metrics declared pre-pilot
- Use FMCSA + state-of-MN SOFI to anchor target lists (no synthetic prospects)
- Build trust through audit substrate first (rate-floor guard + immutable event log)

7 · Sources Cited

- American Trucking Associations · Trucking Industry Profile (annual)

- ATRI · An Analysis of the Operational Costs of Trucking · 2024
- FMCSA · Motor Carrier Safety Progress Report
- FMCSA · SAFER company snapshot · <https://safer.fmcsa.dot.gov>
- US Energy Information Administration · weekly retail diesel · <https://eia.gov>
- Bureau of Labor Statistics · 53-3032 Heavy and Tractor-Trailer Truck Drivers · OEWS
- Minnesota DOT · Statewide Freight Plan · <https://www.dot.state.mn.us>
- Trucking Association of Minnesota · <https://mntruck.org>
- McKinsey & Company · Travel, Logistics and Infrastructure insights

— end of report · sources cited inline —